# 1.7 Equilibria

## Question Paper

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|---|---|
| Course | CIE A Level Chemistry |
| Section | 1. Physical Chemistry |
| Topic | 1.7 Equilibria |
| Difficulty | Easy |

**Time allowed:** 20

**Score:** /10

**Percentage:** /100

## Question 1

Dinitrogen tetraoxide, $N_2O_4$, and nitrogen dioxide exist in equilibrium. This is represented by the reaction below:

$$2NO_2\,(g) \rightleftharpoons N_2O_4\,(g) \quad \Delta H = -57\ kJ\ mol^{-1}$$

Which conditions give the greatest percentage of $NO_2$ at equilibrium?

A. High pressure and high temperature

B. Low pressure and high temperature

C. High pressure and low temperature

D. Low pressure and low temperature

**[1 mark]**

## Question 2

Sulphuric acid is manufactured via a series of reactions in the Contact process. The reaction for the main reaction is shown below.

$$2SO_2\,(g) + O_2\,(g) \rightleftharpoons 2SO_3\,(g) \quad \Delta H = -197\ kJ\ mol^{-1}$$

Which statement about the reaction is correct?

A. Iron is used as a catalyst.

B. Increased temperature gives a higher yield of $SO_2$.

C. Increased temperature gives a higher yield of $SO_3$.

D. Increased pressure gives a higher yield of $SO_2$.

**[1 mark]**

## Question 3

Which statement below best describes a dynamic equilibrium?

A. The rate of the forward reaction is equal to the backwards reaction, and the concentrations of reactants and products is constant.

B. The rate of the forward reaction is equal to the backwards reaction in a closed system, and the concentrations of the reactants and products are equal.

C. The rate of the forward reaction is equal to the backwards reaction in a closed system, and the concentrations of the reactants and products is constant.

D. The rate of reaction changes in either direction to counteract a change in conditions.

**[1 mark]**

## Question 4

An equilibrium is set up in a closed container between equal volumes of gaseous reactants A and B to form a gaseous product C.

The total pressure within the container at 50℃ is 3 atm.

$$A (g) + B (g) \rightleftharpoons 2C (g)$$

The equilibrium partial pressure of A at 50℃ is 0.5 atm

What is the partial pressure of C at this temperature?

A. 2.5

B. 2.75

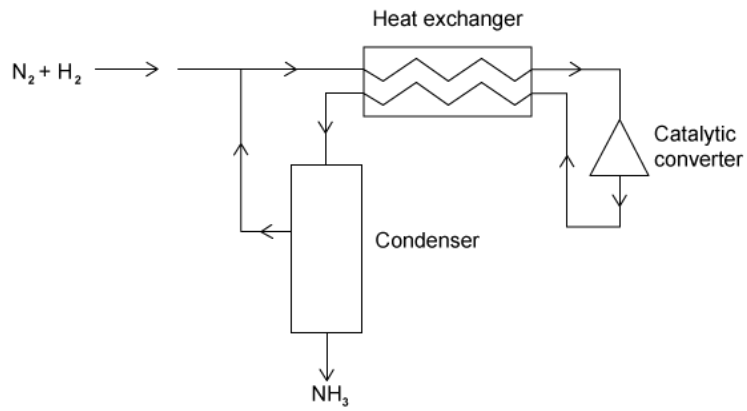
C. 1

D. 2

**[1 mark]**

## Question 5

The Haber process is used to produce ammonia from nitrogen and hydrogen on an industrial scale. The diagram below represents the Haber process.




What is the purpose of the heat exchanger?

A. To cool the incoming gas mixture to avoid overheating the catalyst.

B. To cool the reaction products and separate the ammonia from unused nitrogen and hydrogen.

C. To warm the incoming gas mixture to get more ammonia.

D. To cool the incoming gas mixture and speed up reaction.

**[1 mark]**

## Question 6

Ethanol is manufactured by reacting steam with ethene.

$$C_2H_4\,(g) + H_2O\,(g) \rightleftharpoons C_2H_5OH\,(g) \quad \Delta H = -45\ \text{kJ mol}^{-1}$$

What would increase the equilibrium yield of ethanol in this process?

1. Adding a catalyst.

2. Increasing the pressure.

3. Decreasing the temperature.

A. 1 and 2

B. 1 and 3

C. 2 and 3

D. 1, 2 and 3

**[1 mark]**

## Question 7

Propyl ethanoate is formed in an esterification reaction. How can the value of the equilibrium constant $K_c$ be increased?

$$\text{Propan-1-ol} + \text{ethanoic acid} \rightleftharpoons \text{propyl ethanoate} + \text{water} \quad \Delta H = -10\ \text{kJ mol}^{-1}$$

A. Increasing the temperature

B. Adding a catalyst

C. Increasing the pressure

D. Lowering the temperature

**[1 mark]**

## Question 8

In the reaction where gaseous iodine reacts with hydrogen, an equilibrium is established at 450℃. The reaction is exothermic.

$$H_2(g) \quad + \quad I_2(g) \quad \rightleftharpoons \quad 2HI(g)$$

| colourless | purple | colourless |
|---|---|---|

Which change in conditions will cause the purple colour of the equilibrium mixture to become paler?

A. Decrease in pressure

B. Decrease in temperature

C. Increase in pressure

D. Increase in temperature

**[1 mark]**

## Question 9

For the equilibrium $2A(g) + B(g) \rightleftharpoons 2C(g)$ $\Delta H = +6.5\ \text{kJ mol}^{-1}$, what will change the value of $K_p$?

A. Adding a catalyst

B. Decreasing the pressure

C. Increasing the pressure

D. Increasing the temperature

**[1 mark]**

## Question 10

An equilibrium is established in the reaction.

$$AB\,(aq) + CD\,(aq) \rightleftharpoons AC\,(aq) + BD\,(aq) \quad \Delta H = +180\ kJ\ mol^{-1}$$

Which factors would affect the value of $K_c$ in this equilibrium?

A. Change in temperature in the absence of a catalyst

B. Change in pressure in the presence of a catalyst

C. Increasing the concentration of AB (aq)

D. Increasing the concentration of AC (aq)

**[1 mark]**

# Model Answers: Easy

1

The correct answer is **B** because:

- The **enthalpy change** of the forward reaction is **exothermic** as the sign of ΔH **is negative.**
- Therefore, the **backward** reaction is **endothermic.**
- In accordance with **Le Chatelier's principle**, the equilibrium will shift to the endothermic side of the reaction (the reactant side of the reaction) when the temperature is raised to **oppose** the change.
- This has the effect of i**ncreasing** the **percentage of $NO_2$** in the equilibrium mixture.
- In accordance with Le Chatelier's principle, the equilibrium will shift to the side with **more moles** of a gas when pressure is **lowered** to oppose the change.
- There are more moles of gas on the left (2 moles on the reactant side, 1 mole on the product side). Therefore, the equilibrium will shift to the left-hand side of the reaction when pressure is lowered to oppose the change.
- This has the effect of **increasing** the percentage of $NO_2$ in the equilibrium mixture.

**A** is incorrect as high pressure would shift the equilibrium to the side of the reaction with fewer moles of a gas. For this reaction, the equilibrium will thus shift to the right-hand side of the reaction in accordance with Le Chatelier's principle. A low temperature would shift the equilibrium to the right-hand (exothermic) side.

**C** is incorrect as whilst this is a favourable **temperature** for a **high yield** of $NO_2$, high pressure would shift the equilibrium to the side with fewer moles of a gas, the equilibrium will thus be shifted to the right-hand side of the reaction.

**D** is incorrect as both low pressure and low temperature would shift the equilibrium to the right-hand side of the reaction in accordance with Le Chatelier's principle.

2

The correct answer is **B** because:

- The enthalpy change of the forward reaction is **exothermic** as the sign of ΔH is **negative.**
- Therefore, the backwards reaction is **endothermic.**
- The equilibrium will shift to the **left** (the reactant side of the reaction) when the temperature is raised to **oppose** the change.
- This has the effect of **increasing the yield** of $SO_2$ in the equilibrium mixture.
- This is in accordance with **Le Chatelier's** principle.

**A** is incorrect as iron is used in the **Haber process,** not the Contact process. The Contact process uses **vanadium oxide (V)** as its catalyst.

**C** is incorrect as **increasing the temperature** would shift the equilibrium to the **endothermic**, backward, reaction which would lower the yield of $SO_3$.

**D** is incorrect as increasing the pressure will shift the equilibrium the side with less moles of a gas. This would shift the equilibrium to the right-hand (product) side of the reaction as there are fewer moles of gas (3 on the reactant side and 2 on the product side) in accordance with Le Chatelier's principle.

3

The correct answer is **C** because:

- The three key ideas in a dynamic (moving) equilibrium is that:
  1. Rate of the forward reaction = rate of backward reaction.
  2. The reaction is in a closed system (meaning nothing can escape).
  3. The concentration of reactants and products are constant. This can be in a 50%- 50% equilibrium yield, 75 %- 25%, 60%-40 etc. The key point is that they remain unchanged in that distribution once equilibrium has been reached.

**A** is incorrect as they have not mentioned the fact that is must be in a closed system. The definition is, therefore, incomplete.

**B** is incorrect as they have stated that '*The concentrations of the reactants and products are equal*'. This is not true. They are **constant**, not equal.

**D** is incorrect as this is describing a shift in equilibrium, not the features of dynamic equilibrium.

4

The correct answer is **D** because:

- As there are equal volumes of reactants A and B in a 1:1 molar ratio means their partial pressures will be the same. B therefore also has an equilibrium partial pressure of 0.5.
- Remember Total Pressure = ∑ equilibrium partial pressures.
- Therefore adding all the partial pressure together must = 3.
- 0.5+ 0.5 + *p(C) = 3*
- Therefore *p(C)* = 2 atm.

**A** & **B** are incorrect as adding up the partial pressures here would exceed the total pressure which does not agree with the equation Total Pressure = ∑ equilibrium partial pressures.

**C** is incorrect as adding up the partial pressures here would equal total pressure of 2 atm. Again, this does not agree with the equation Total Pressure = ∑ equilibrium partial pressures.

5

The correct answer is **C** because:

- The **Haber process** reacts nitrogen and hydrogen at a high temperature and pressure to produce ammonia in the reaction: $N_2 + 3H_2 \rightleftharpoons 2NH_3$.
- Ammonia is used as a fertiliser in agriculture.
- Increasing the temperature gives molecules more **kinetic energy**; therefore the gases **collide more frequently**, which increases the likelihood of a reaction.

- More collisions enable the reaction to reach equilibrium faster, and therefore more ammonia is produced than if an equilibrium isn't reached.

- The **Haber process** reacts nitrogen and hydrogen at a high temperature and pressure to produce ammonia in the reaction: $N_2 + 3H_2 \rightleftharpoons 2NH_3$.
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- Increasing the temperature gives molecules more **kinetic energy**; therefore the gases **collide more frequently**, which increases the likelihood of a reaction.
- More collisions enable the reaction to reach equilibrium faster, and therefore more ammonia is produced than if an equilibrium isn't reached.

The forward reaction, producing ammonia, is exothermic (it produces heat) and therefore, according to Le Chatelier's principal will be favoured at lower temperatures. The warming of the gas mixture focuses on the forward reaction **before** equilibrium is reached. At a low temperature, the gases won't have enough kinetic energy to collide and react, and therefore equilibrium would not be reached, and the amount of ammonia produced would be very low.

**A** is incorrect as the catalyst will not overheat, the Haber process takes place at 400°C.

**B** is incorrect as ammonia is easily liquified under pressure when the gases leave the reactor.

**D** is incorrect as cooling the gases will reduce the rate of reaction.

6

The correct answer is **C** because:

- In accordance with **Le Chatelier's principle, high pressure** would shift the equilibrium to the side of the reaction with **fewer moles** of a gas.
- In this reaction, the equilibrium would therefore shift to the right-hand side of the reaction.
- This would **increase** the yield of ethanol.
- The **enthalpy change** of the forward reaction is **exothermic** as the sign of ΔH **is negative.**
- In this reaction, when lowering the temperature, the equilibrium would shift to the

right-hand side of the reaction to counteract the change, which is the exothermic direction.

- This would **increase** the yield of ethanol.

**A, B** and **D** are incorrect as adding a catalyst will increase the **rate** of the reaction meaning the state of equilibrium will be reached faster but will have **no effect on the position of the equilibrium once equilibrium is reached.**

7

The correct answer is **D** because:

- The enthalpy change of the forward reaction is exothermic. Therefore the backwards reaction is endothermic.
- The equilibrium will shift in the right (the product side of the reaction) when the temperature is lowered to oppose the change.
- This has the effect of increasing the yield of propyl ethanoate in the equilibrium mixture. This is in accordance with Le Chatelier's principle.
- When applying this change to the equation for $K_c$ the numerator (the top bit of the equation) has increased, therefore the values of $K_c$ increases.
- $K_c = \dfrac{[propyl\ ethanoate][water]}{[ethanoic\ acid][propan-1-ol]}$

**A** is incorrect as the equilibrium will shift to the left, lowering the yield of propyl ethanoate, in accordance with Le Chatelier's principle. This will increase the denominator (bottom bit) of the $K_c$ equation, lowering its value.

**B** is incorrect as adding a catalyst will increase the **rate** of the reaction meaning the state of equilibrium will be reached faster but will have **no effect on the position of the equilibrium.**

**C** is incorrect as pressure only affects species in the gaseous state.  None of the reactants or products is gaseous in this reaction. **Only a temperature can change the equilibrium constant $K_c$, not changes in pressure, and or a catalyst.**

8

The correct answer is **B** because:

- The enthalpy change of the forward reaction is exothermic. Therefore the backwards reaction is endothermic.
- The equilibrium will shift in the right (the product side of the reaction) when the temperature is lowered to oppose the change.
- This has the effect of increasing the yield of HI in the equilibrium mixture causing the solution to become paler in colour.
    - This is in accordance with Le Chatelier's principle.

**A** & **C** are incorrect as in accordance with Le Chatelier's principle, low pressure would shift the equilibrium to the side of the reaction with more moles of a gas and vice versa if the pressure is increased. However, in this reaction, there are the **same moles of gas on each side.** Therefore there is no change in the position of the equilibrium.

**D** is incorrect as increasing the temperature will increase the **rate** of the **endothermic** (backwards) reaction meaning the colour of the solution will get darker, not lighter.

9

The correct answer is **D** because:

- Only temperature permanently affects the value of $K_p$ as the equilibrium will shift to oppose the change.
- No other change in condition will affect the value of $K_p$.

**A** is incorrect as adding a catalyst will increase the **rate** of the reaction meaning the state of equilibrium will be reached faster but will have **no effect on the position of the equilibrium.** Therefore the value of $K_p$ is unchanged.

**B** & **C** are incorrect as **only temperature will affect the value of $K_p$. IMPORTANT NOTE: DON'T MENTION LE CHATELIERS PRINCIPLE HERE.** When pressure is changed, it's important to remember that the pressure will affect all the gases equally as they are in the same closed container of fixed volume. However, when you apply the same change to the numerical value to each gas' partial pressure in the $K_p$ equation, the value of $K_p$ will change (only for an instant) as they have different exponents. The equilibrium will shift to oppose this change in order to keep $K_p$ constant. E.g. if you raise the pressure in this example. The value of the denominator (the bottom bit of the equation) will increase to a greater extent as they have the greater exponent (the small number raised to the power). This will cause the equilibrium to shift to the right to increase the numerator (the top bit of the equation) and so the value of $K_p$ is maintained.

9

The correct answer is **D** because:

- Only temperature permanently affects the value of $K_p$ as the equilibrium will shift to oppose the change.
- No other change in condition will affect the value of $K_p$.

**A** is incorrect as adding a catalyst will increase the **rate** of the reaction meaning the state of equilibrium will be reached faster but will have **no effect on the position of the equilibrium.** Therefore the value of $K_p$ is unchanged.

**B** & **C** are incorrect as **only temperature will affect the value of $K_p$. IMPORTANT NOTE: DON'T MENTION LE CHATELIERS PRINCIPLE HERE.** When pressure is changed, it's important to remember that the pressure will affect all the gases equally as they are in the same closed container of fixed volume. However, when you apply the same change to the numerical value to each gas' partial pressure in the $K_p$ equation, the value of $K_p$ will change (only for an instant) as they have different exponents. The equilibrium will shift to oppose this change in order to keep $K_p$ constant. E.g. if you raise the pressure in this example. The value of the denominator (the bottom bit of the equation) will increase to a greater extent as they have the greater exponent (the small number raised to the power). This will cause the equilibrium to shift to the right to increase the numerator (the top bit of the equation) and so the value of $K_p$ is maintained.

10

The correct answer is **A** because:

- Remember **only temperature** will affect the value of the $K_c$.
- Any other change in condition would result in $K_c$ staying constant.

**B** is incorrect as a change in pressure can only effect gases. Take a look at the state symbols; they are all aqueous. Adding a catalyst will increase the **rate** of the reaction meaning the state of equilibrium will be reached faster but will have **no effect on the position of the equilibrium.** Therefore the value of $K_c$ is unchanged.

**C** & **D** are incorrect as by changing the concentration of either reactant or product, the equilibrium will shift in such a way that **the ratio of concentration to product remains the same.** Therefore $K_c$ is unchanged.